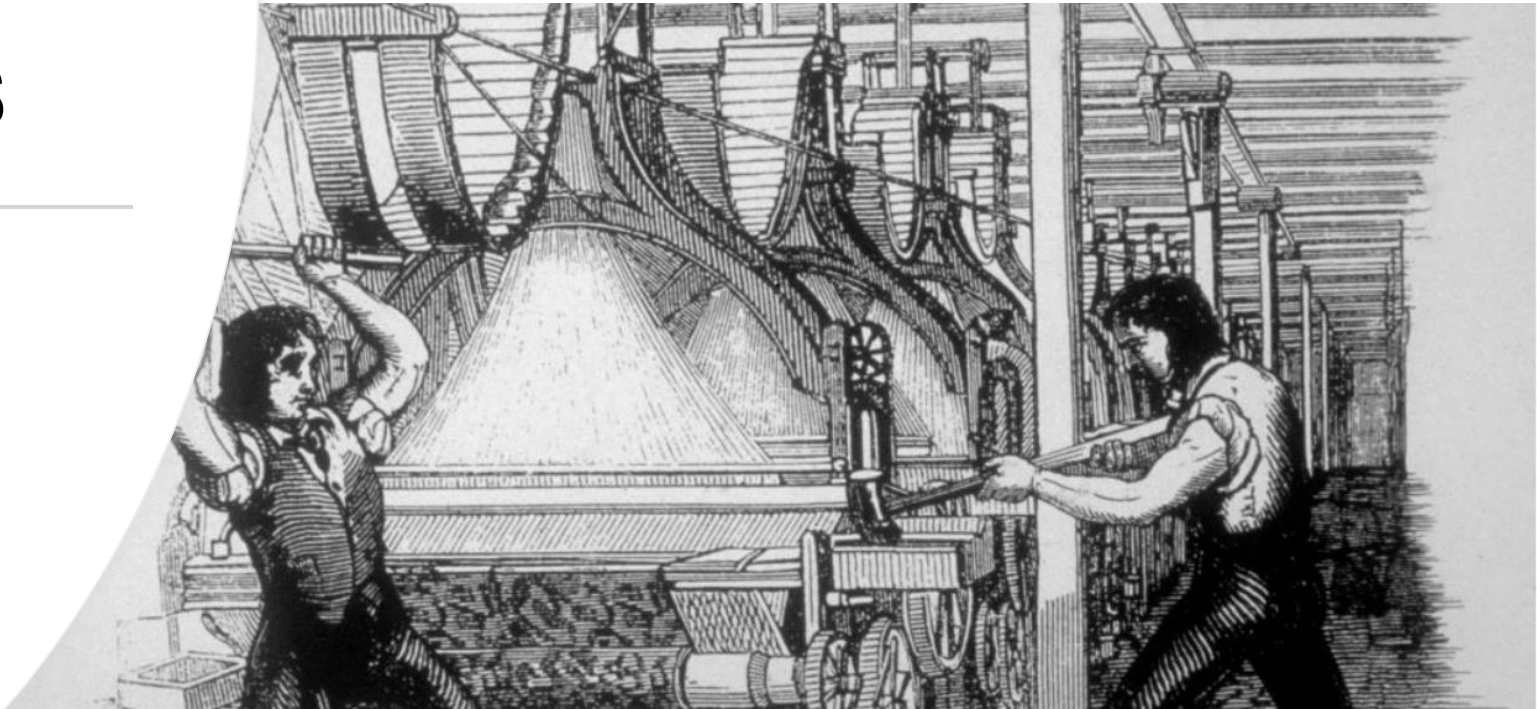
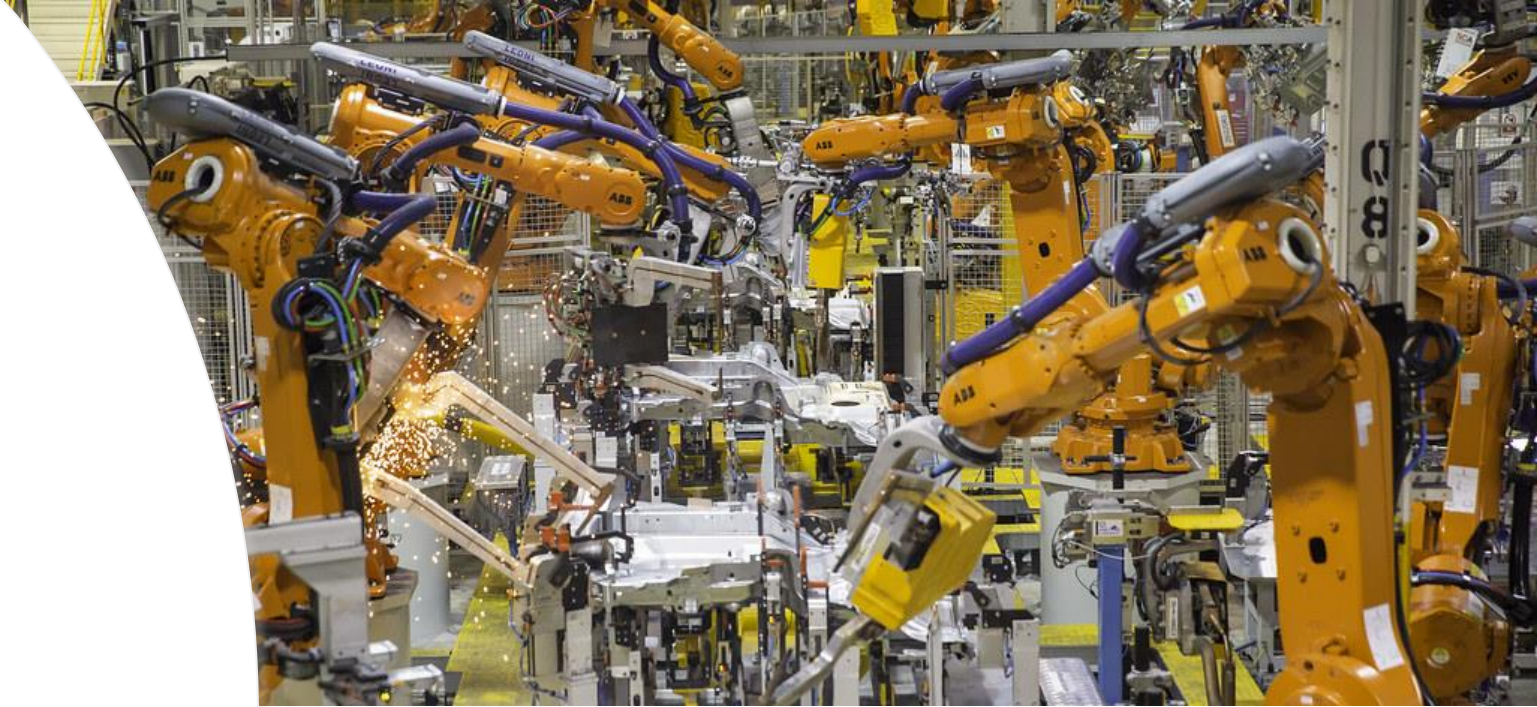


Econ 139: Labor Economics

Big Questions in Labor Economics

David Arnold



Welcome to Econ 139: Labor Economics

- Today's class
 - Go over course structure
 - Discuss Big Questions in Labor Economics

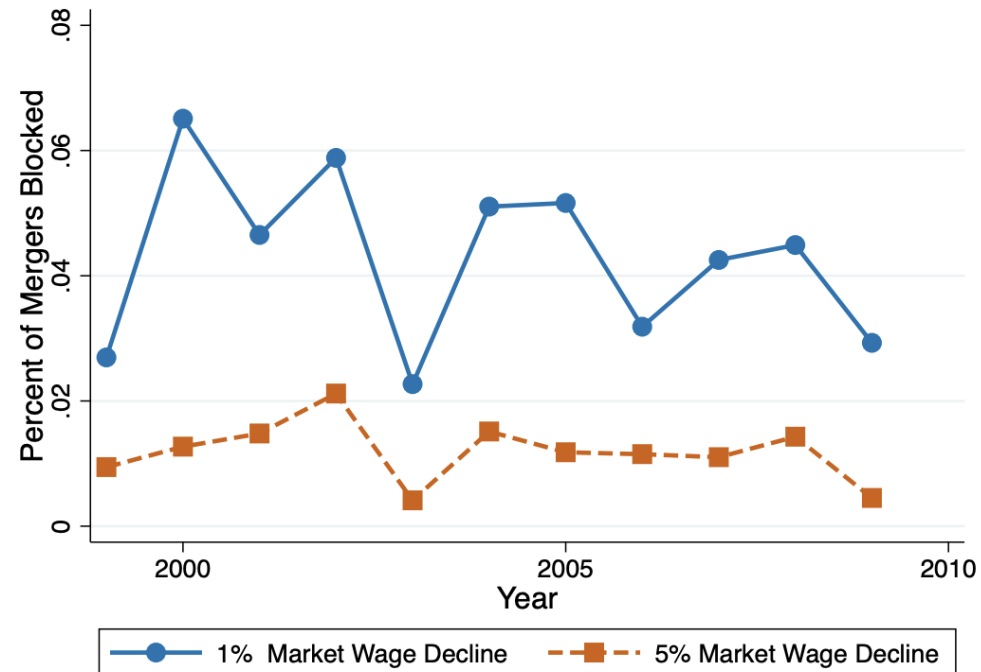
Introductions

- David Arnold
 - PhD from Princeton in 2020
- My research focus is in two very different strands of labor economics

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Area 1: imperfect competition in labor markets. For example, do mergers have negative impacts on workers due to anticompetitive effects?

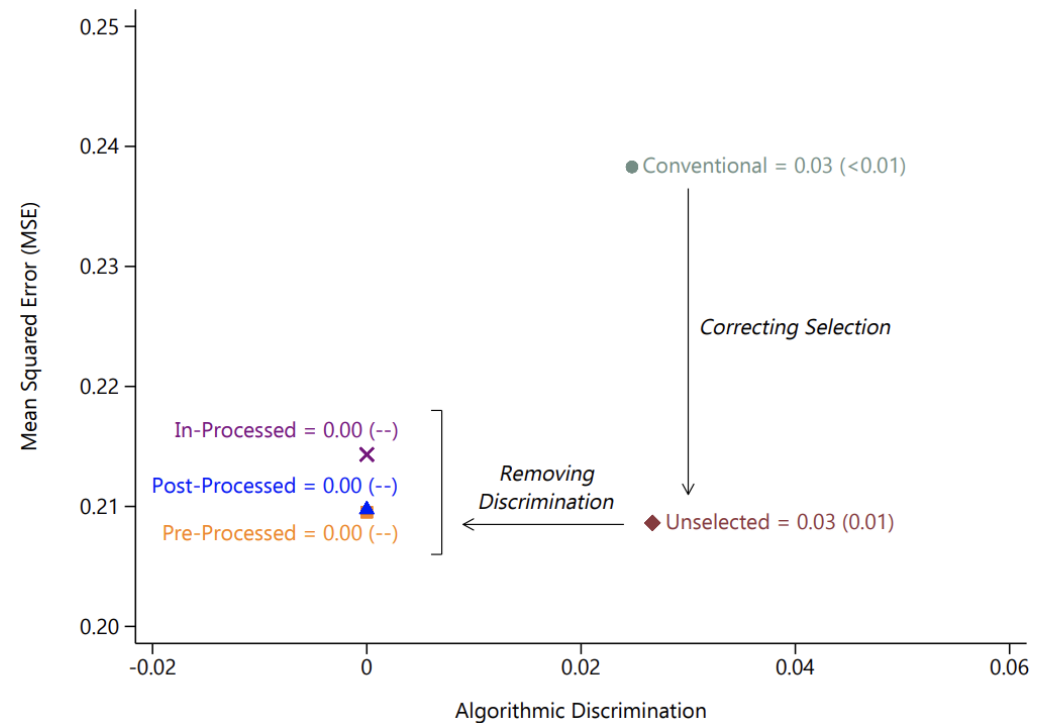


Introductions

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Area 2: Identifying racial discrimination in judges and algorithms

Figure 3: Algorithmic Discrimination and Accuracy



What is Labor Economics?

- In this class, we will study what determines the employment and wage level in the labor market
 - The labor market is not like any other market. Workers have preferences. If you are a firm, you can't just go out and find the perfect worker for your job. The models we learn in introductory courses may not apply
 - It is one of the most important institutions in modern society. Changes in the labor market have profound impacts on people's lives, politics, and society at large

What is Labor Economics?

- One way to understand labor economics is through exploring some of the biggest questions facing labor economists
- We will dig deeper into these questions at the end of this lecture
- First we are going to go over some administrative matters.



Course Materials

- There is an online textbook that closely follows the slides. You don't need to buy any materials for the class, but if you'd like suggestions for further reading, feel free to ask
- As part of the class, we will analyze data. The software will be made available through canvas
 - software is generally not compatible with chromebooks or ipads (see the Syllabus if you don't have a computer that is compatible with the software)

Example of a Normal Week



- About 2/3 of material will be standard lectures on either theoretical models of labor market and studying prior work
- About 1/3 of material be utilizing labor-market data. It will be helpful to bring a laptop to these classes.
- Each week on Canvas there is a “Plan of the Week” that will release slides, show deadlines for assignments, assign readings and label which classes will be data-focused

Grade Breakdown

- Participation (5%) – I will take attendance in class. If you attend 70% or more of the classes, you will receive 100 percent
- Problem Sets (5% each, 15% total) – There will be 3 problem sets. For each, you will have a week and a half to complete
- Reflections (5% total) – Short reflections on reading assignments
- Midterm (20%) – In class on Wednesday October 29th.
- Presentation (20%) – More on this below
- Final (35%) – Tuesday December 9th at RWAC 0121, 11:30 AM-2:30 PM

Midterm and Final

- You must be available to take the exams in person to participate in this class. **Exceptions are made only for verifiable medical, legal or sports reasons.**
- If you do not attend an exam without a verified reason then you will receive a zero on the exam.

Other Administrative Matters

- There will be no class on either Wednesday November 26th or Friday November 28th
- Make sure to take survey with #FinAid on Canvas

Presentation

- Each student will choose an occupation
 - The presentation will present trends in employment and wages, discuss tasks common in the job, and forecast potential future market implications, all using data we discuss in class
- You will record you presentation on zoom and then upload to Canvas (i.e. it will not be an in person presentation)
- As part of the assignment, you will be placed in small groups to give feedback on other students' presentations



Academic Integrity (Problem Sets and Presentation)

- You can use AI to help with problem sets and presentations
 - The problem sets have been written in a way in which AI may be useful, but cannot completely do the assignment for you. Because we are allowing the use of AI, incorrect information or illogical statements will receive harsh penalties in terms of grading (i.e. just copy and pasting the results from an LLM is likely to result in a low grade)
- You can collaborate with other students on problem sets, but what you write down should be your own work (i.e. you cannot copy and paste or simply put your name on someone else's work. If two assignments are too similar, you may be reported to Academic Integrity office)

Academic Integrity (Midterm and Final)

- The midterm and final will be in-person. There will be no use of outside materials for either exam. Any suspected violation will be submitted to the Academic Integrity Office for an investigation

Big Questions in Labor Economics

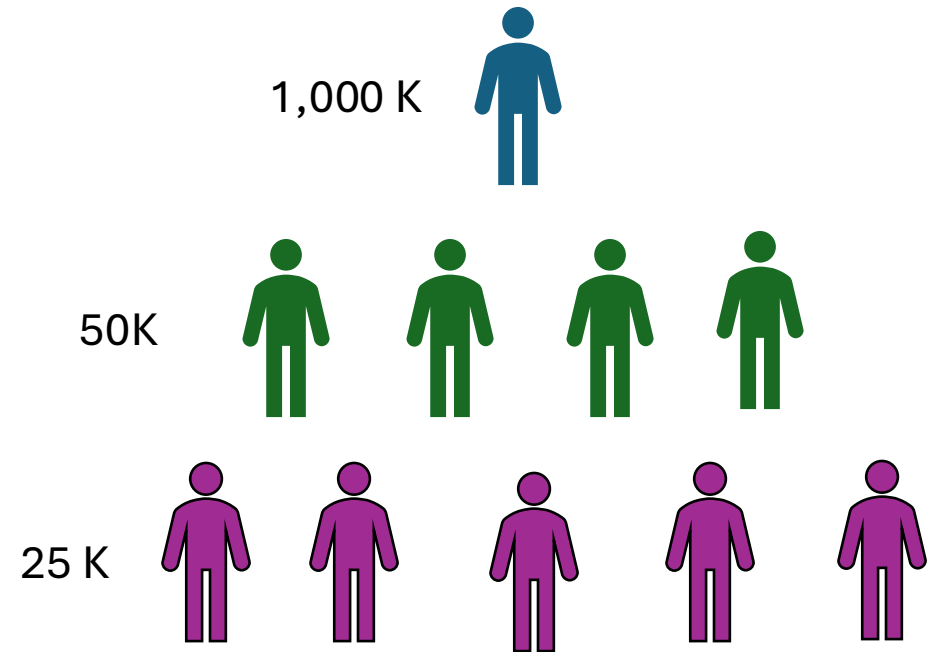
- To guide this class, we are going to periodically link back to three big questions in Labor Economics
 - **Question 1:** The Rise in Inequality
 - **Question 2:** The Fall of the Labor Share
 - **Question 3:** Dramatic Shifts in Labor Force Participation

Rising Inequality

- Inequality is one of the most frequently discussed aspects of the labor market
- But when people discuss inequality, they often have different measures of inequality
 - There is not one single measure of inequality that everyone agrees on
- In this section, we are going to go through a couple of different measures of inequality

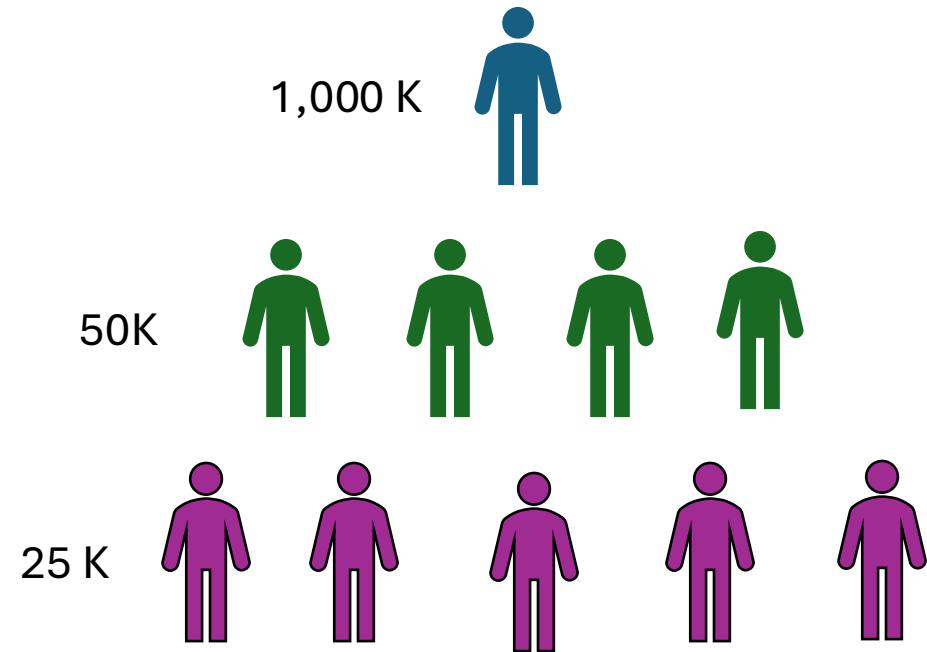
Our first measure of inequality

- Imagine we have 10 households
- 1 household makes 1 million a year
- 4 make 50,000 a year
- 5 make 25,000 a year



Our first measure of inequality

- Let's compute all of the income in the economy
- $1,000,000 + 4(50,000) + 5(25,000) = 1,325,000$
- So the richest person earns about 75% of all income in this example ($1,000,000 / 1,325,000$)



Our first measure of inequality

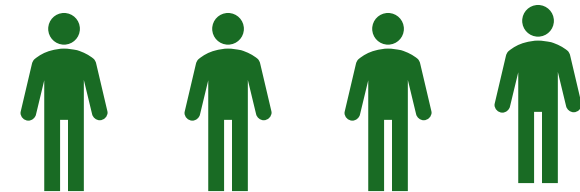
- Since there are 10 people total, the richest person represents the top 10% of all earnings

1,000 K



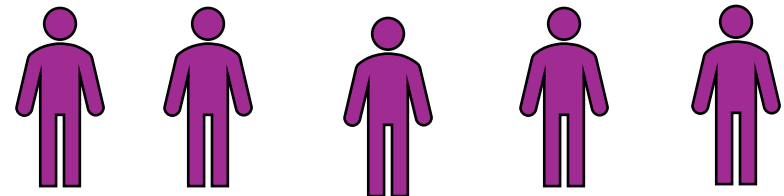
- The bottom 5 represent the bottom 50 percent of all earners

50K



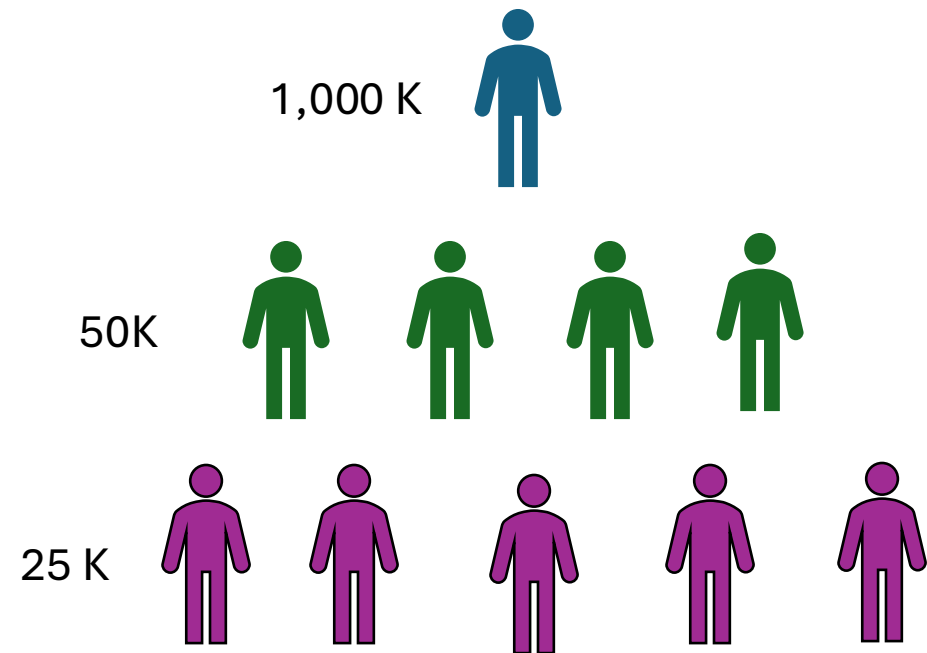
- This bottom 5 earn about 10 percent of all income

25 K



Our first measure of inequality – Shares of Income

- This is a common way to discuss income inequality
- Top 10 percent vs. Bottom 50 percent
- **Top 1 percent vs. Bottom 50 percent**
- Top 0.1 percent vs. Bottom 50 percent





**Type your full name as it appears in
Canvas (this is to take attendance)**

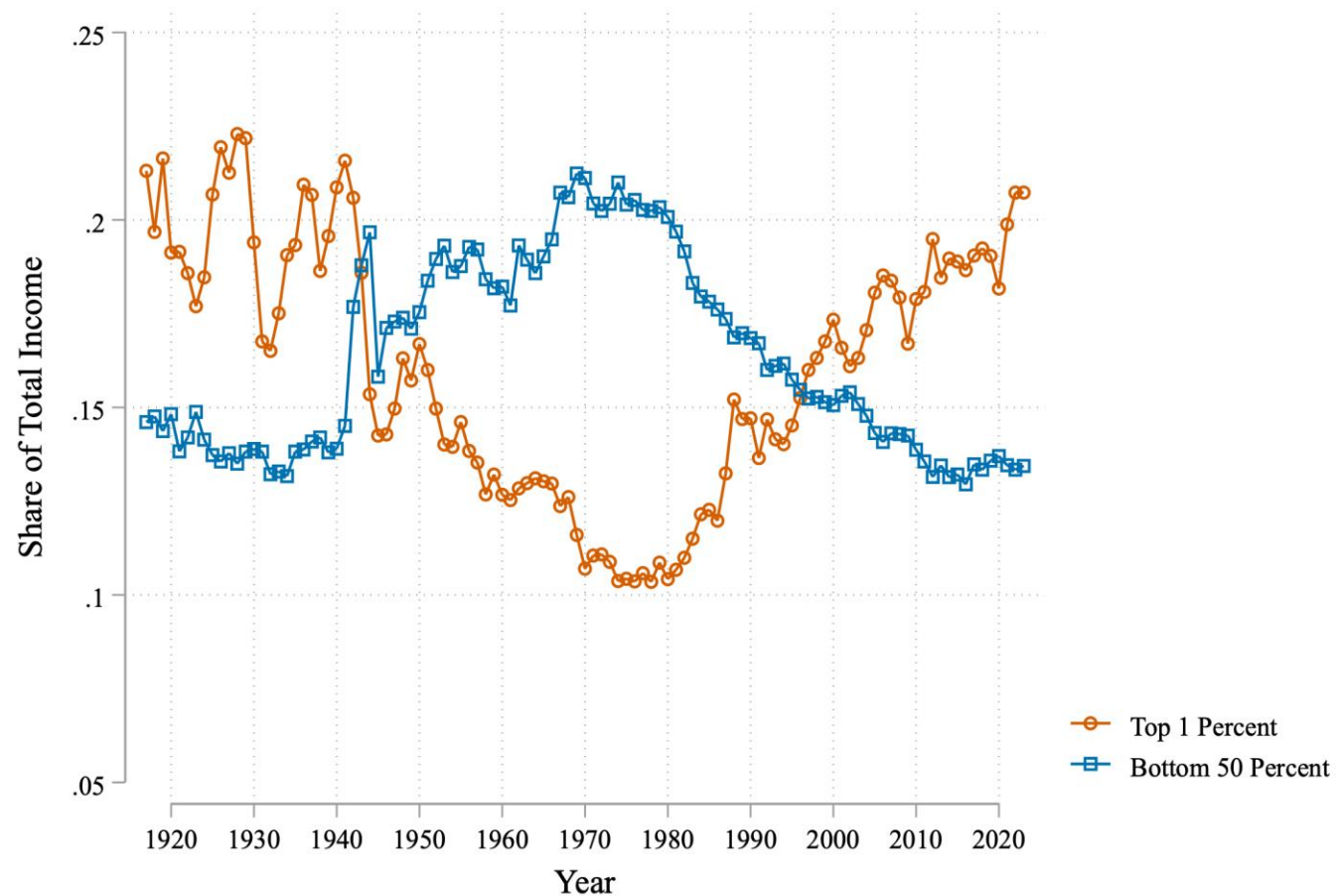


In the 1920s, what percent of income do you think went to the top 1 percent of income earners?



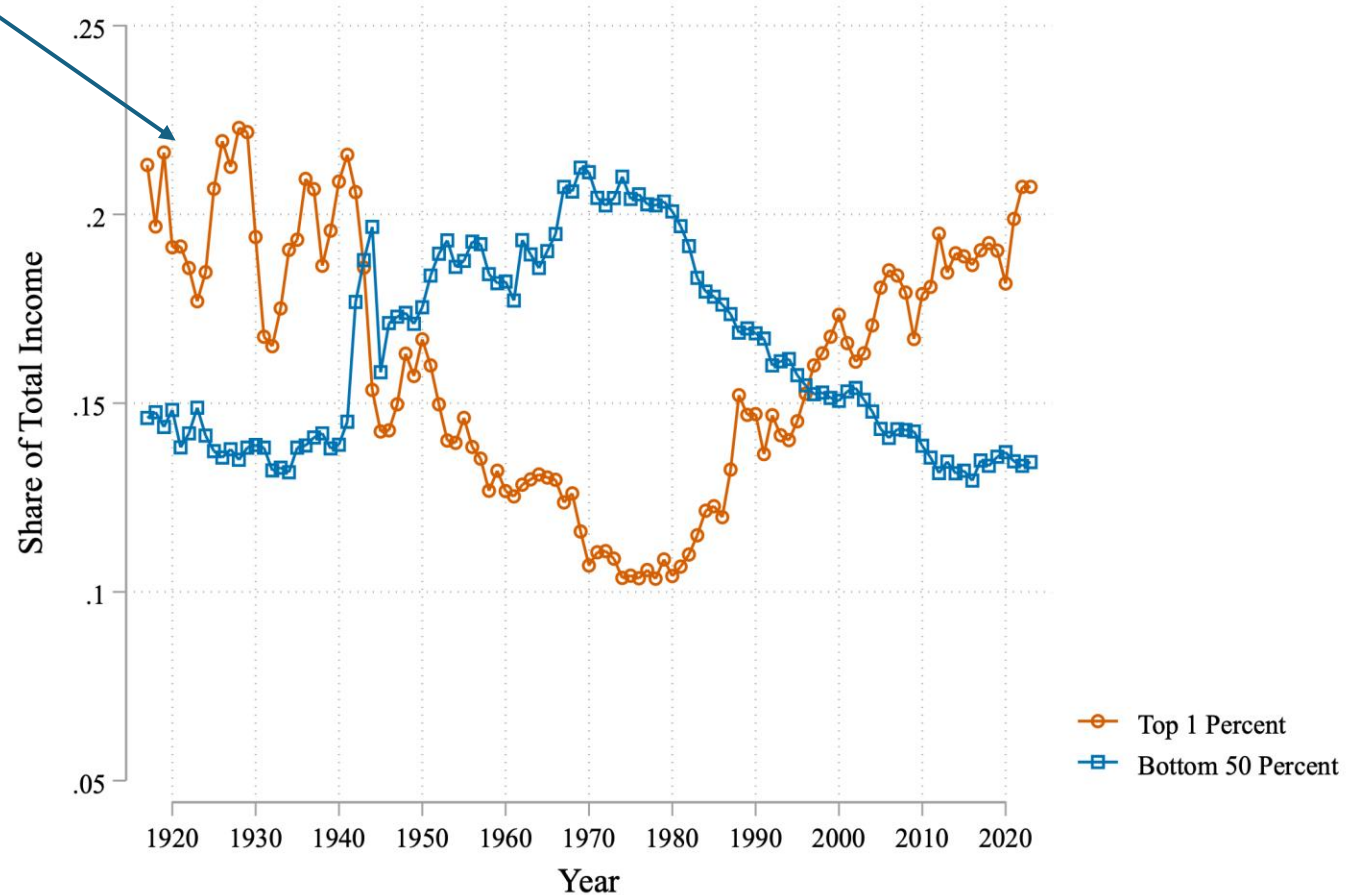
In the 2020s, what percent of income do you think goes to the top 1 percent of income earners

Income Inequality in the 20th Century



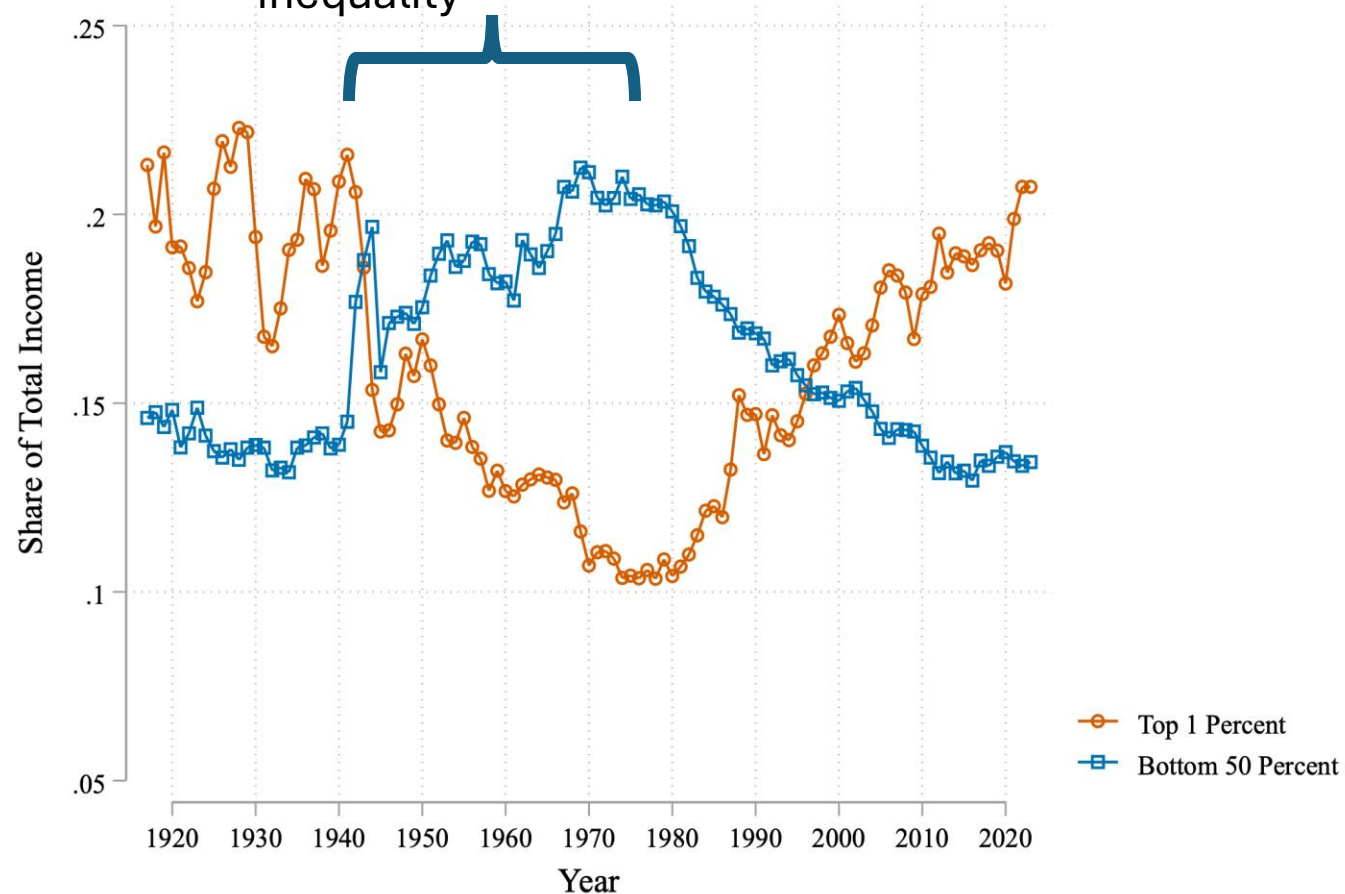
Robber barons – Rockefeller,
Carnegie, Ford, Morgan
amassed massive fortunes

Income Inequality in the 20th Century



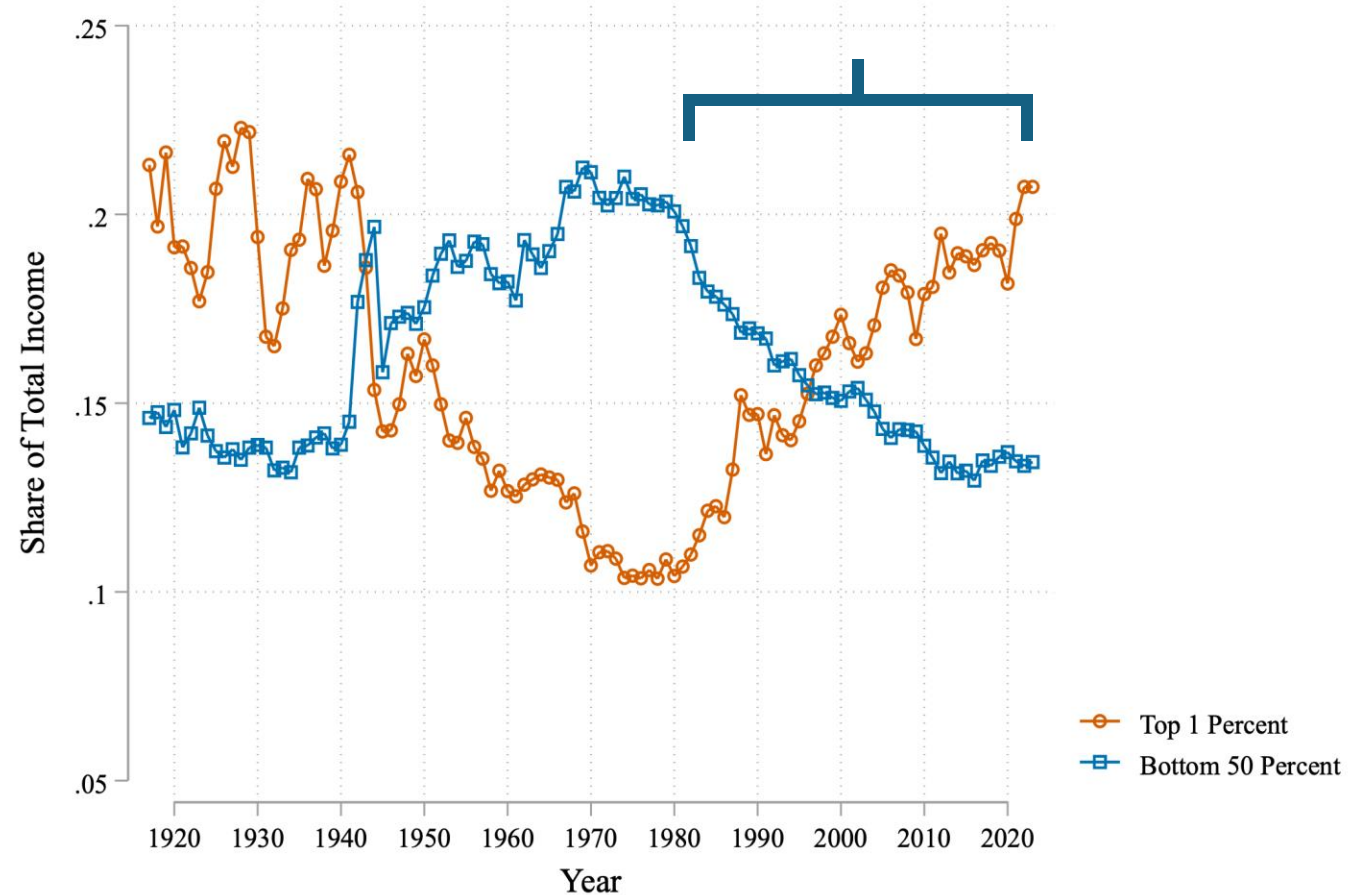
Income Inequality in the 20th Century

The Great Compression -- Post World War II saw rapid expansion of middle class, lowering earnings inequality



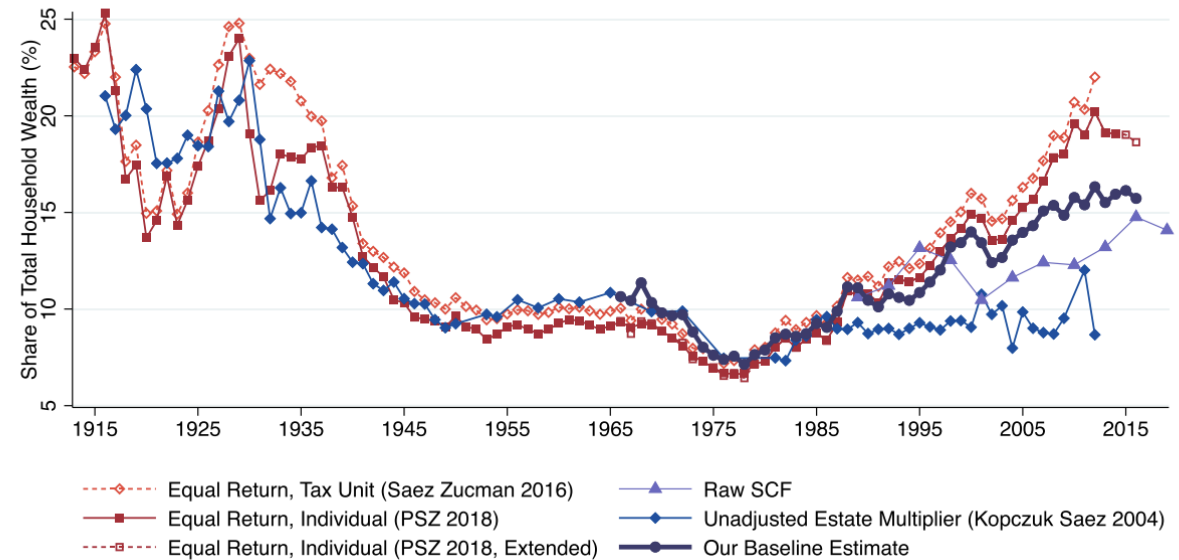
Income Inequality in the 20th Century

The Great Divergence – Beginning
in 1980s, rapid rise in inequality,
back to 1920 levels



Wealth Inequality is Different than Income Inequality

(A) Top 0.1% Share of Total Wealth, Unadjusted Series



Wealth is notoriously difficult to measure, so there are some disputes on exactly how wealth has evolved

The U-Shaped Pattern of Inequality

- The income patterns come from a famous figure in economics from a paper titled *Income Inequality in the United States, 1913-1998*, by Thomas Piketty and Emmanuel Saez
- The numbers are striking with enormous shifts in the fraction of income going to the bottom 50 percent vs. top 1 percent

Our Second Measure of Inequality

- Using shares is useful, but has one important limitation
- It doesn't actually tell us how well-off people are doing
 - Income is not a fixed pie, it can expand. If the economy grows, it is possible that everyone is growing richer over time, even if inequality is expanding
- Next, we are going to turn to a different measure of inequality based on **real income**.

Real Income



- **Real Income** is the amount of income adjusted for inflation
 - If you compare incomes in the 1950s to today, of course earnings were much less in the past. The key, however, is to figure out how much you can afford today vs. in the past
- This is something popular-press or sensationalist headlines can often get wrong
 - Sometimes one good is cherry-picked to try and make the point people were much richer in the past than in the present, or proper adjustments for inflation are outright ignored

Real Income



- The median annual household income in 1950 was about \$3,000, while it was about \$80,000 in 2023
- To understand whether real income has gone up or down, we need a way to compare what we can buy in 2023 with \$80,000 vs. what we can buy in 1950 with \$3,000

Real Income – Getting into the details

- Let's assume that households only pay for two things: housing and groceries.
 - To make it even simpler, let's assume the only grocery is bread
- Prices in 1950 vs 2023

	Housing (Median Rent)	Groceries (Bread)
1950		
2023		

Real Income – Getting into the details

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1950	\$42	
2023	\$1,300	

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$$\text{Price Increase in Housing} = \frac{1300}{42} \approx 30$$

Real Income – Getting into the details

- Let's assume that households only pay for two things: housing and groceries.
 - To make it even simpler, let's assume the only grocery is bread
- Prices in 1950 vs 2023

	Housing (Median Rent)	Groceries (Bread)
1950	\$42	\$0.30
2023	\$1,300	\$2.00

$$\text{Price Increase in Bread} = \frac{2}{0.3} \approx 10$$

Price Index

- A price index aggregates different prices to form an overall price level
- To begin, we often set a base year for the price index (so that values of the price index are relative to the base year)

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- To begin, we often set a base year for the price index (so that values of the price index are relative to the base year)
- For example, imagine I set the base year to 1950 and let $Prices_{1950} = 1$. Then $Prices_{2023}$ tells you how much more expensive prices are in 2023 relative to 1950
- $RealIncome_t = \frac{Income_t}{Price_t}$

Price Index

- Now assume I spend 50% on housing and 50% on groceries, I form the overall price index by multiplying the change in price by the good's consumption share

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$$\bullet \text{ } Prices_{2023} = \underbrace{0.5 \cdot 30} + \underbrace{0.5 \cdot 10} = 20$$

Change in
Housing
Prices

Change in
Grocery
Prices

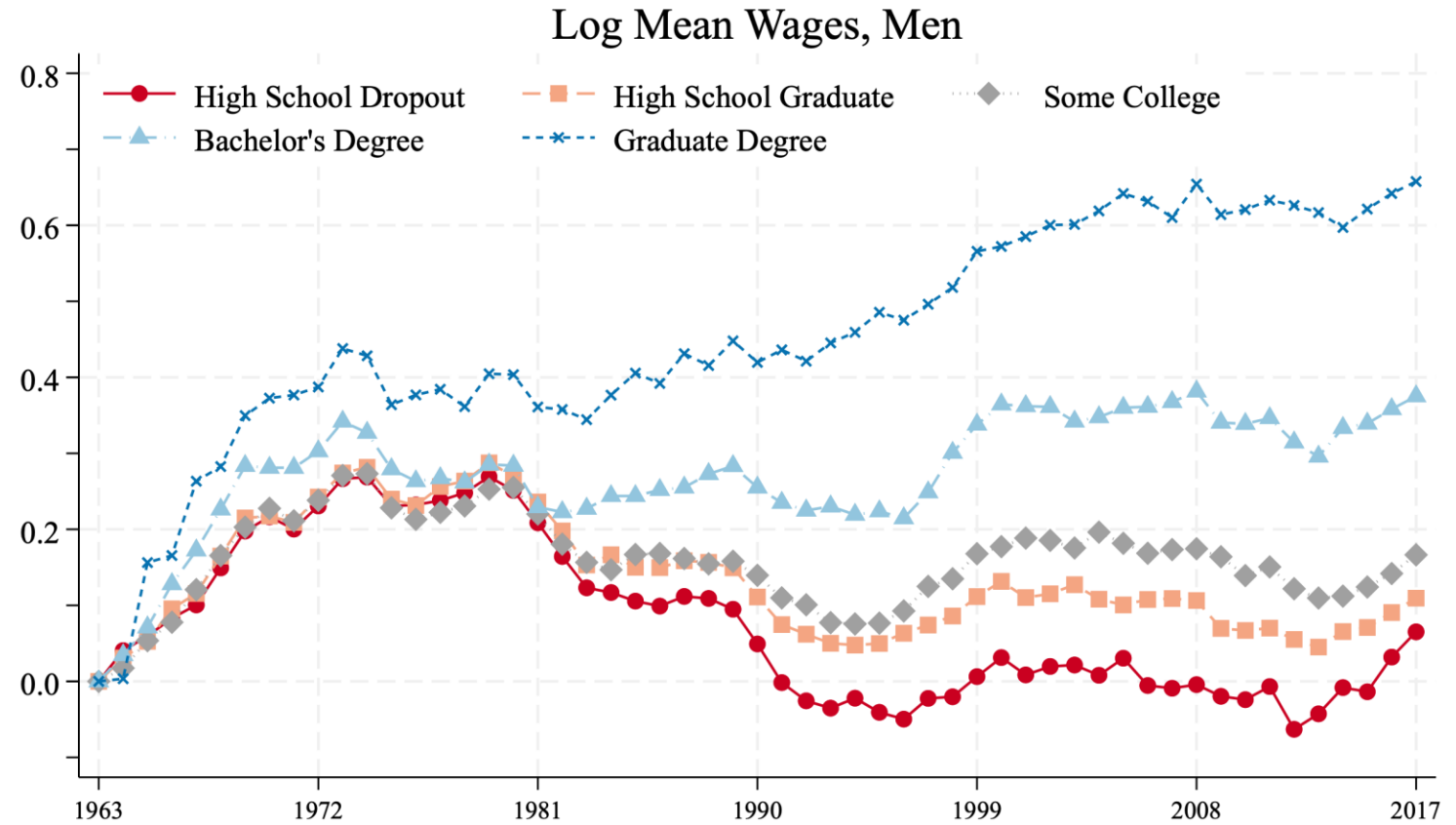
Real Income in 2023

- Remember $RealIncome_t = \frac{Income_t}{Price_t}$
- So for 2023, an income of 80,000 is equivalent to...
- $RealIncome_{2023} = \frac{\$80,000}{20} = \$4,000$, which is greater than the median annual income in 1950, which was \$3,000

Consumer Price Index

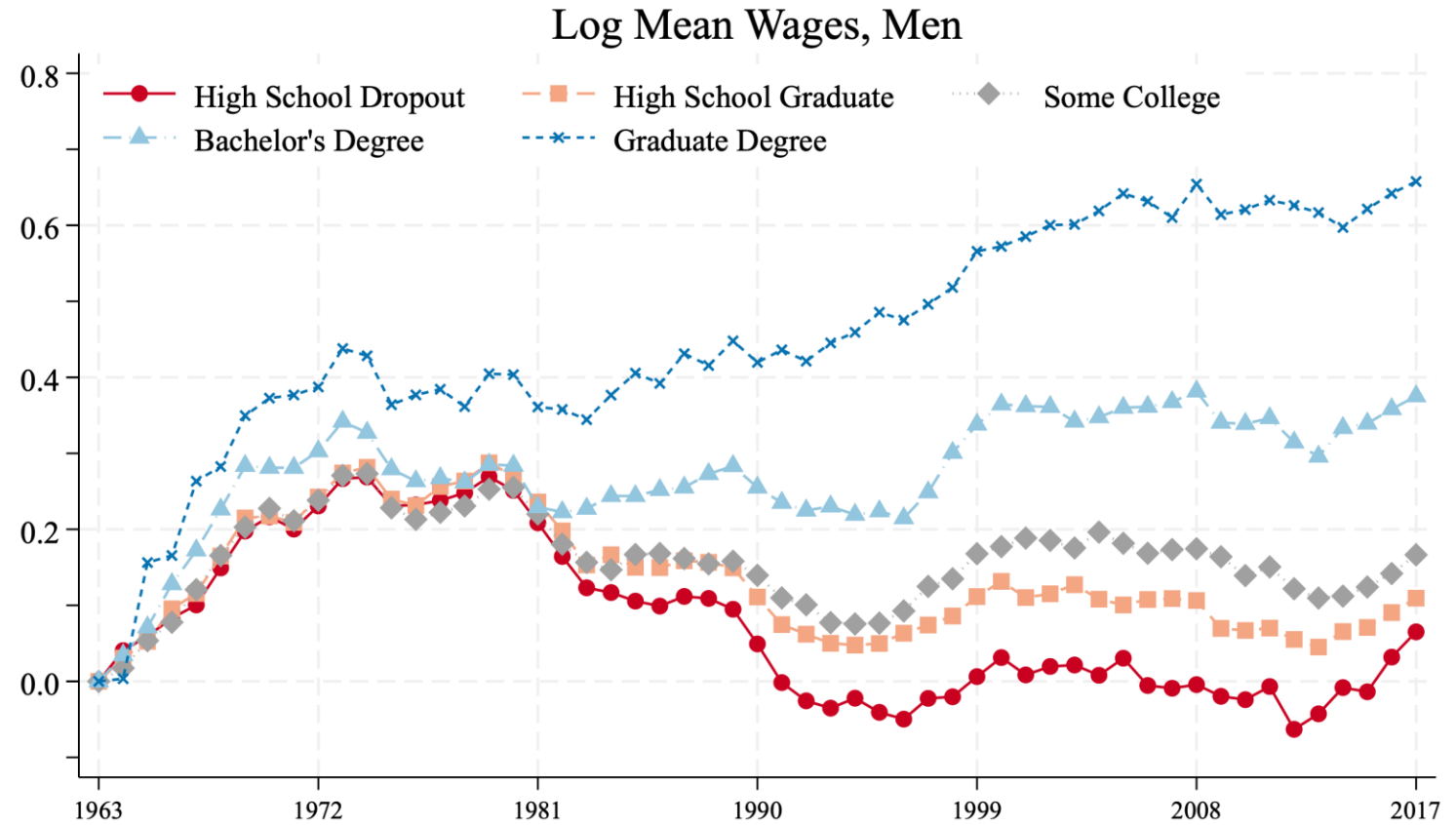
- The Consumer Price Index (CPI) is built on the same foundations here, but instead of two goods, there are thousands
 - There are many, many variants of the CPI that are used for different purposes
- If we want a more accurate picture of how to compare 80,000 today to 3,000, we can directly use the CPI
- $\frac{CPI_{2023}}{CPI_{1950}} = 12.6$, implying \$80,000 is equal to $\frac{\$80,000}{12.6} \approx \$6,350$ in 1950

Real Income Growth by Education



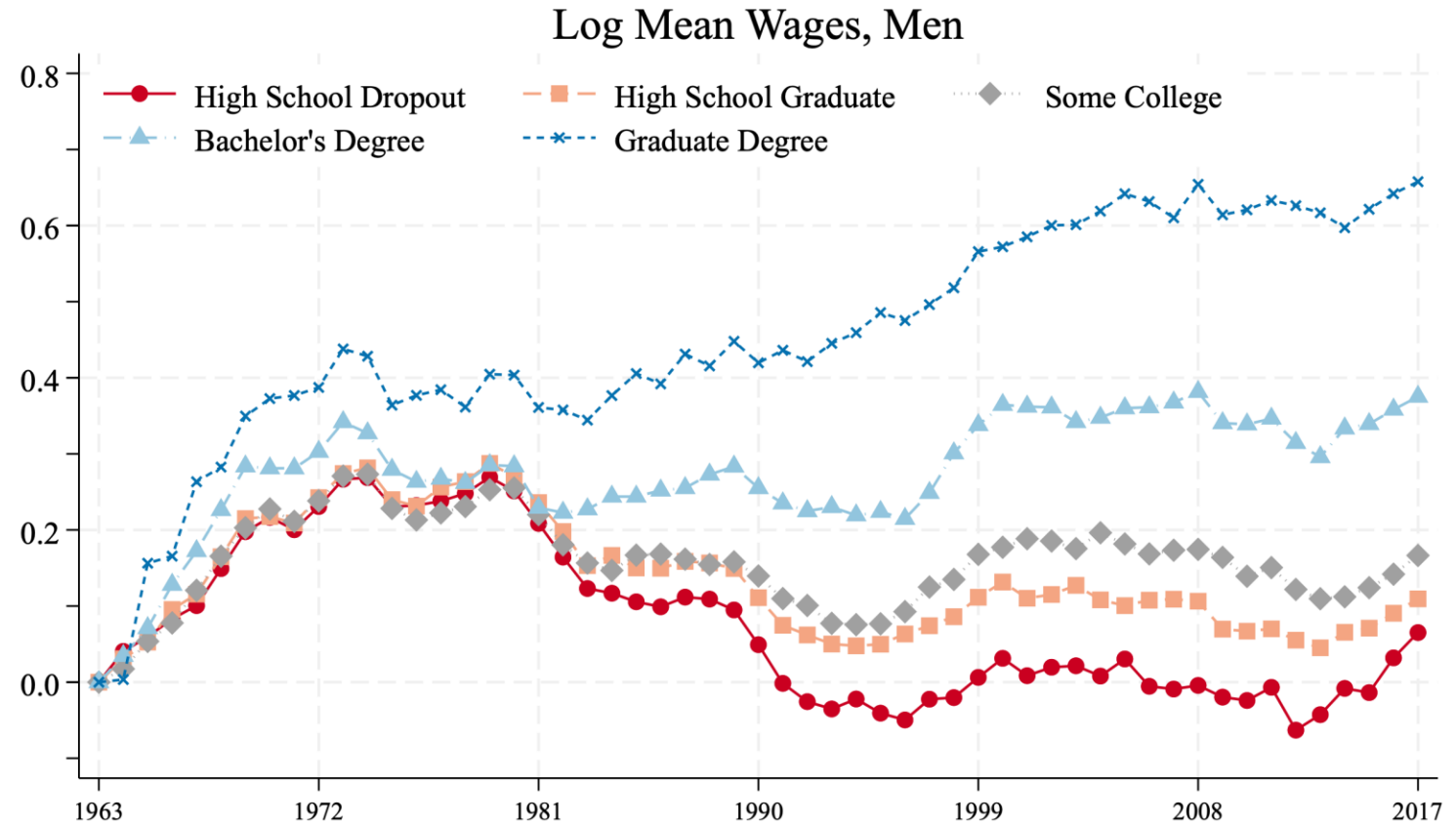
Real Income Growth by Education

- Real income grew rapidly for all workers between 1960 to early 1970s



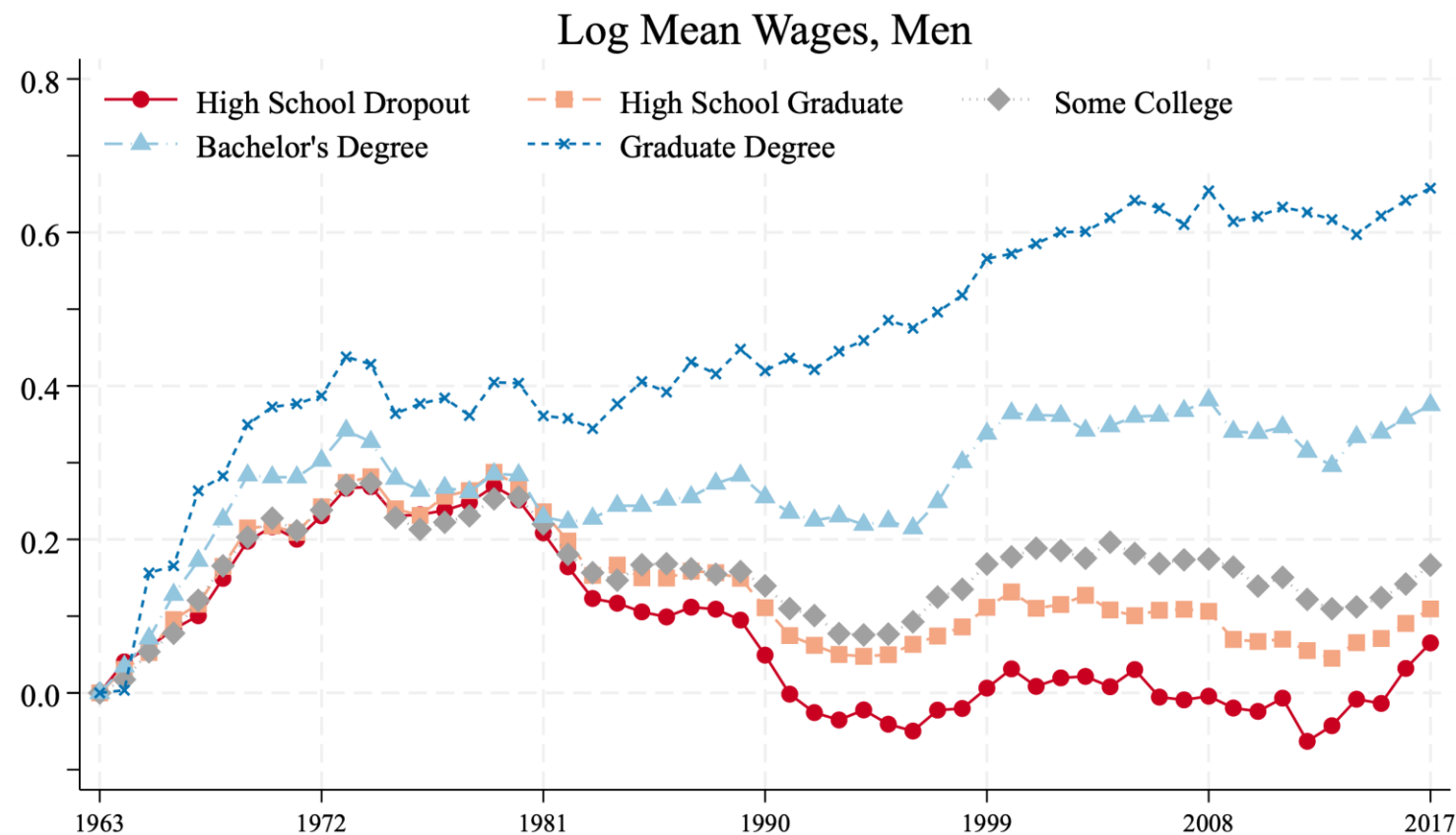
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Real Income Growth by Education

- Real income grew rapidly for all workers between 1960 to early 1970s
- Real income stagnated for most groups in the 1970s
- Real income has fallen for lower-education since the 1980s, but risen for higher-educated groups



Question 2: The Fall in the Labor Share

- Our second big question involves the amount of income going to labor
 - For the inequality portion, we studied how much income goes to different workers. Now, we are studying what fraction of income goes to workers vs. firm owners
- While labor shares can be constructed for the entire economy, we can also construct them for individual firms, which helps understand the concept of a labor share

The Labor Share

- The definition of the labor share is

$$\text{Labor Share} = \frac{\text{Total Employee Compensation}}{\text{Value Added}}$$

- Total Employee Compensation is straightforward, but valued added is a bit trickier to compute
- If a company earns 100K in revenue, but spends 25K on the materials, then the value it added to those materials is 75K
 - In general, we can compute valued added as revenue minus costs (excluding labor costs)

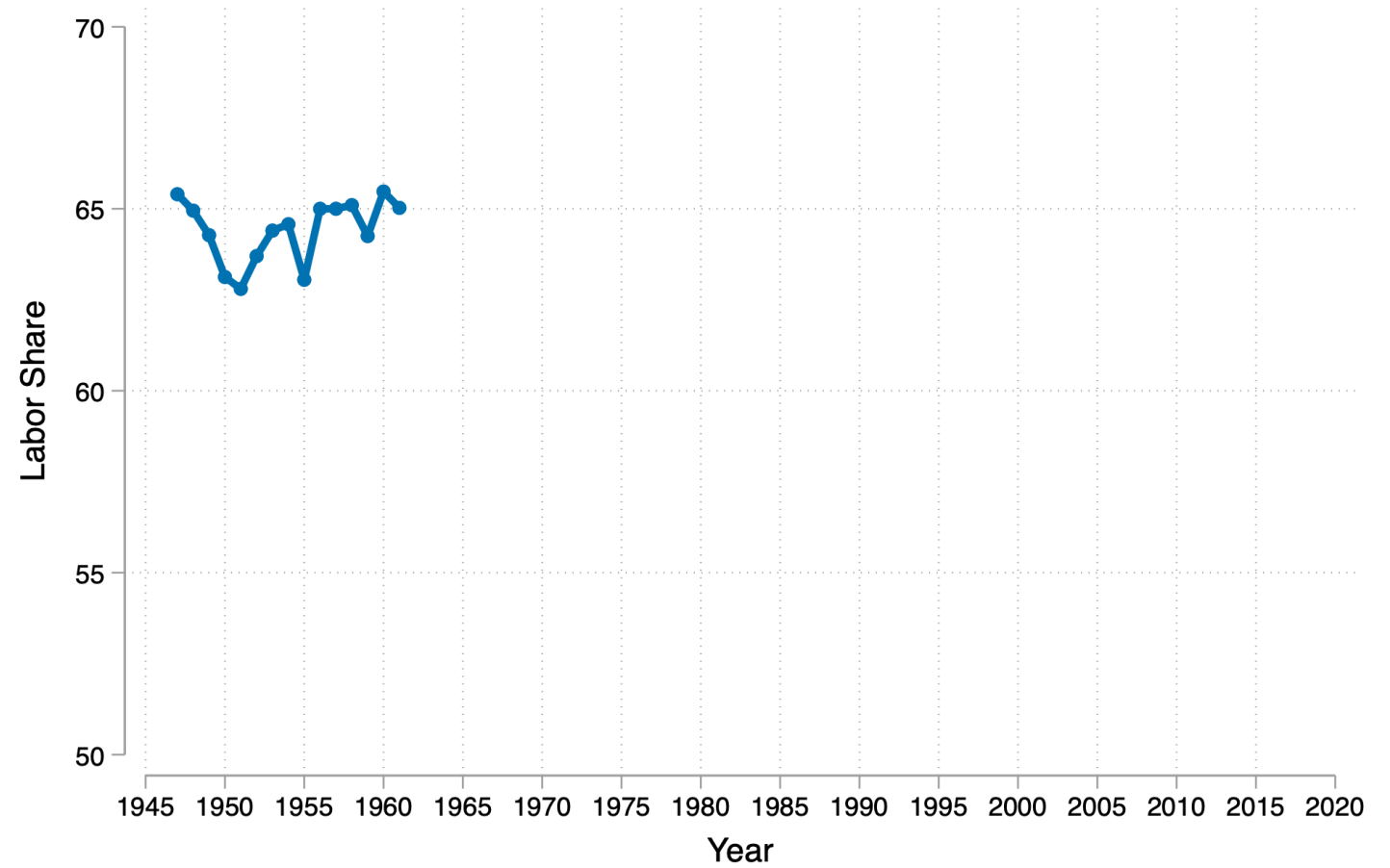
Apple's Labor Share

- In 2024, Apple reported 390 billion in revenue, but they had a large number of other costs, bringing value added to 180 billion
- According to Annual Reports for Apple, workers at apple earned about 46 billion in salaries

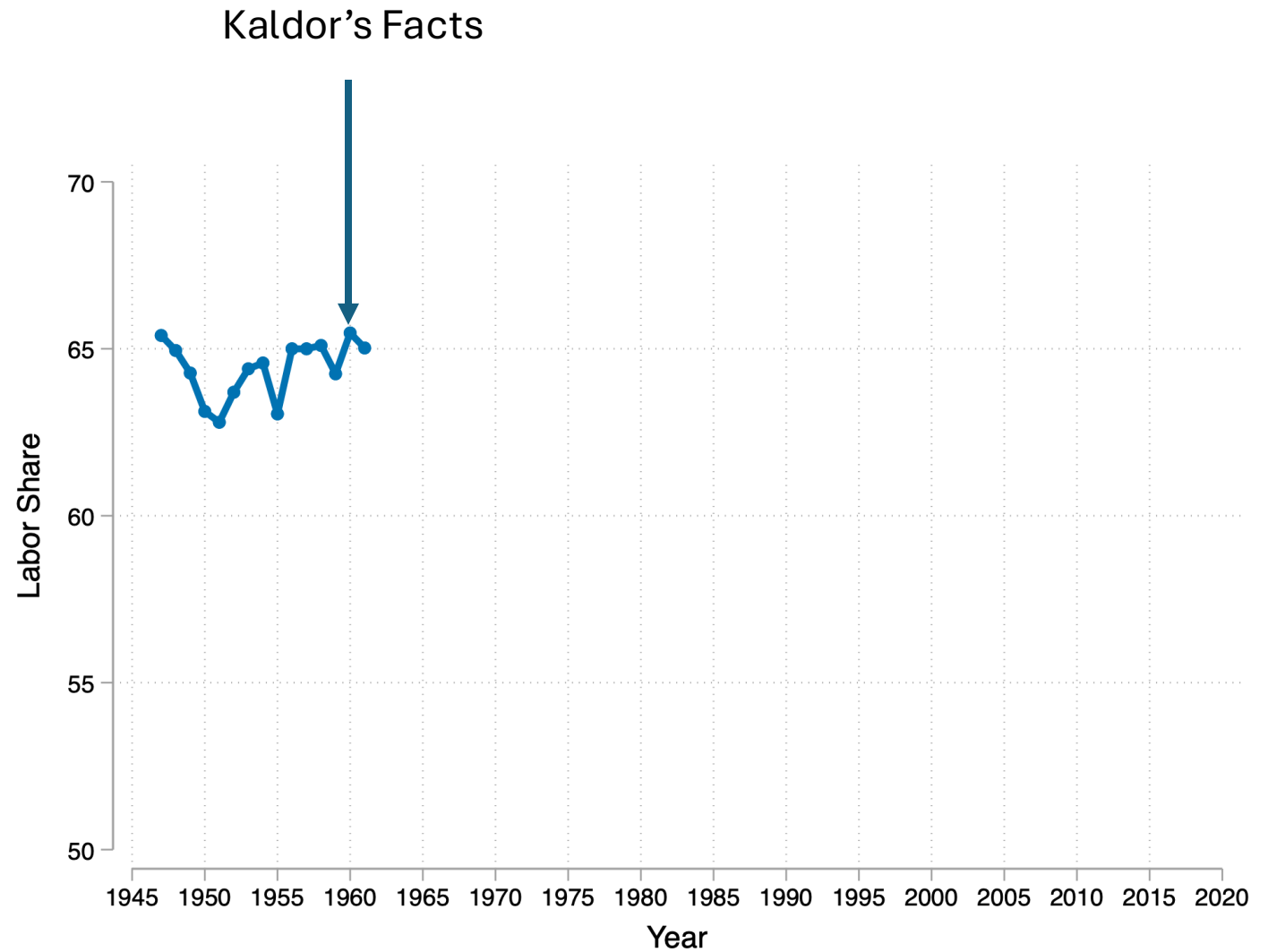
$$\text{Apple Labor Share} = \frac{46}{180} \approx 0.25$$

- So 25% of Apple's value added is paid to workers

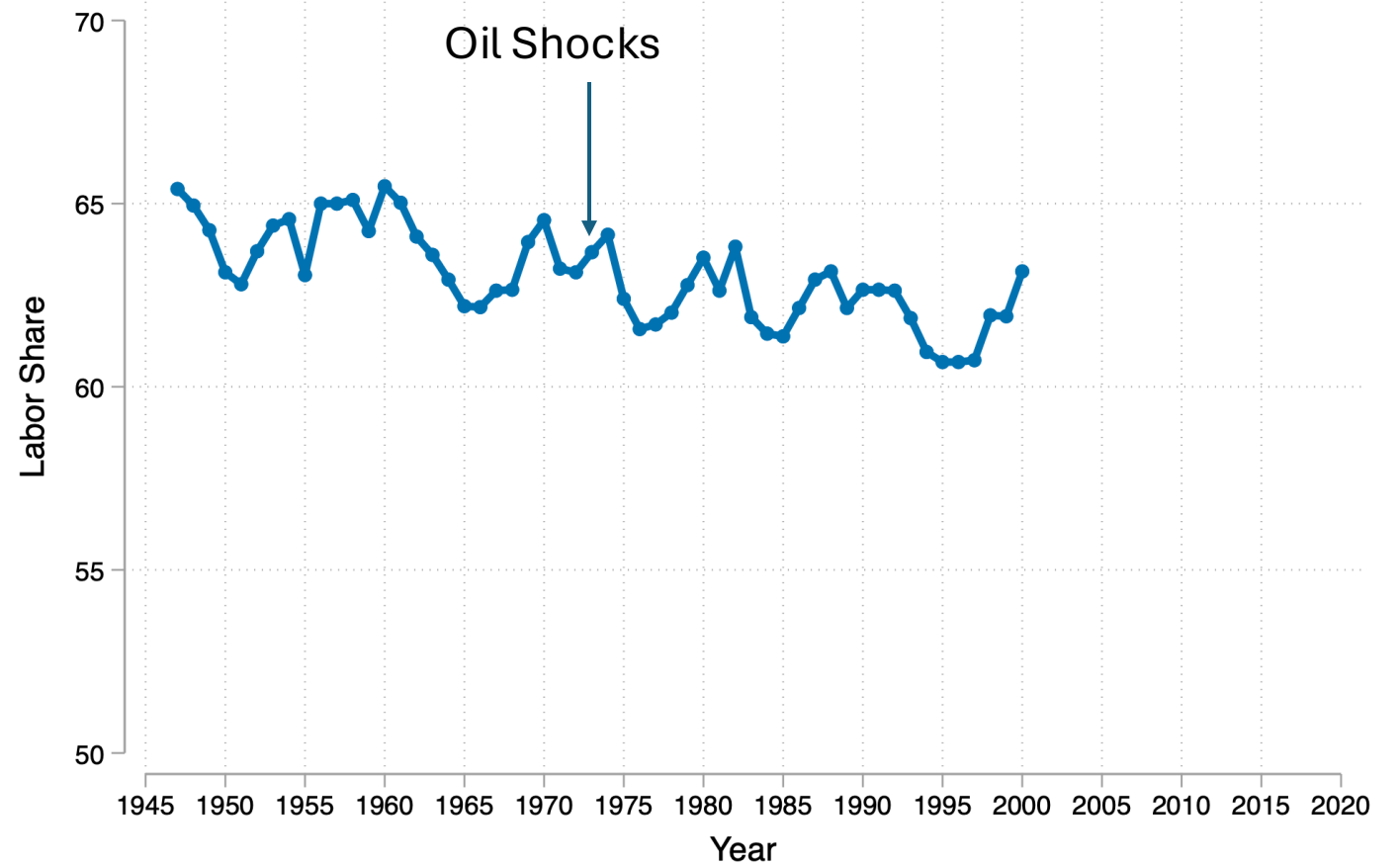
The Labor Share



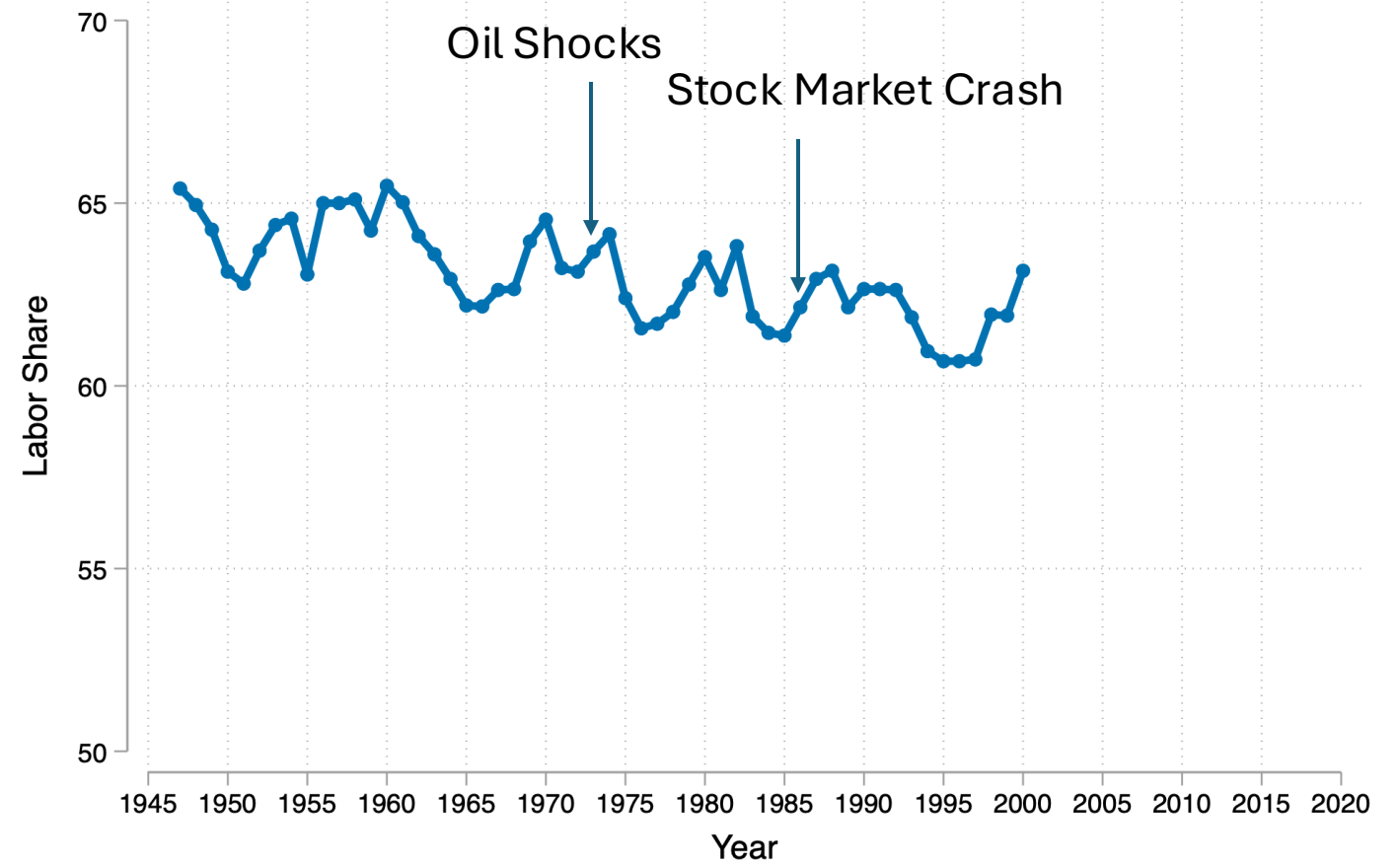
The Labor Share



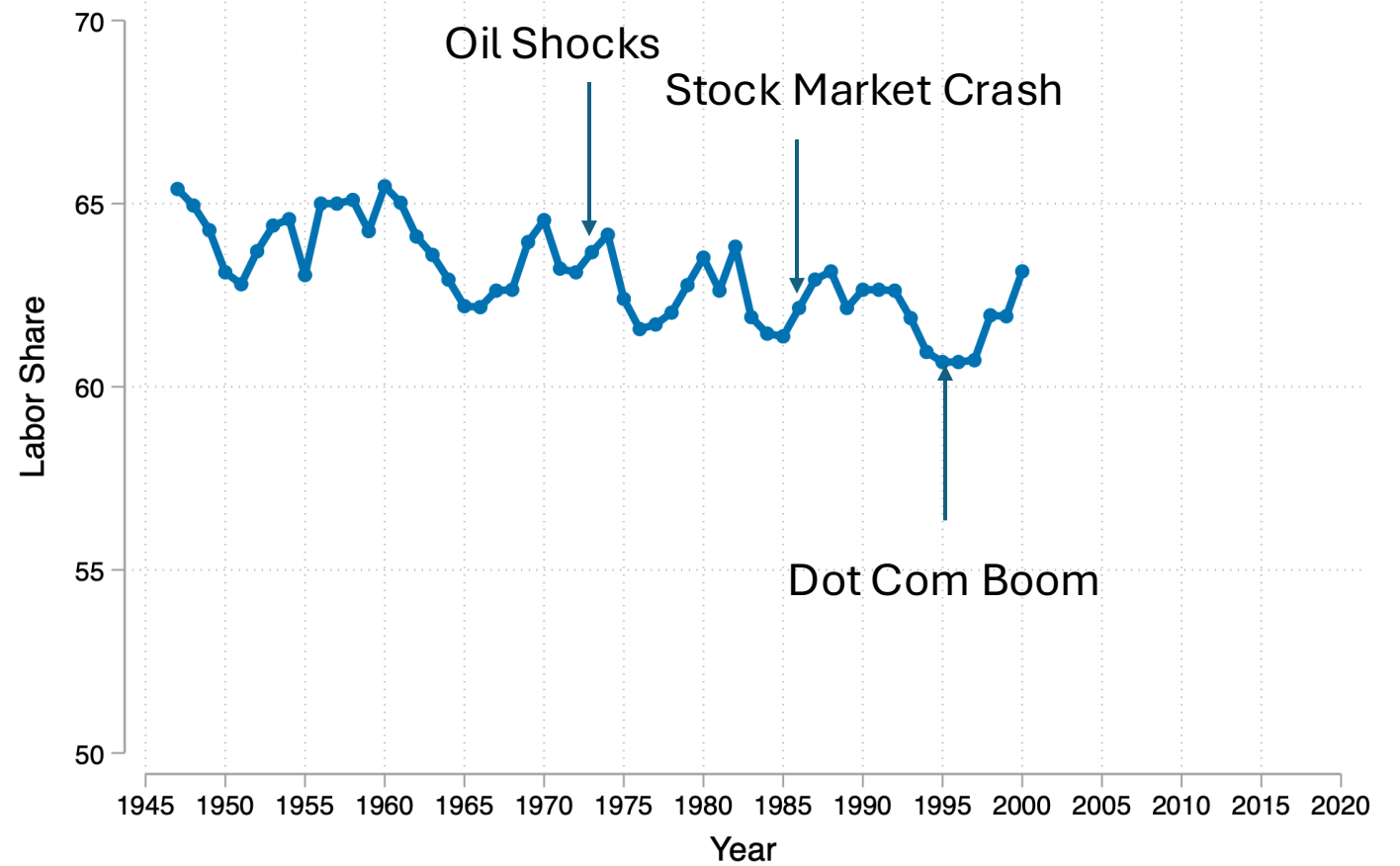
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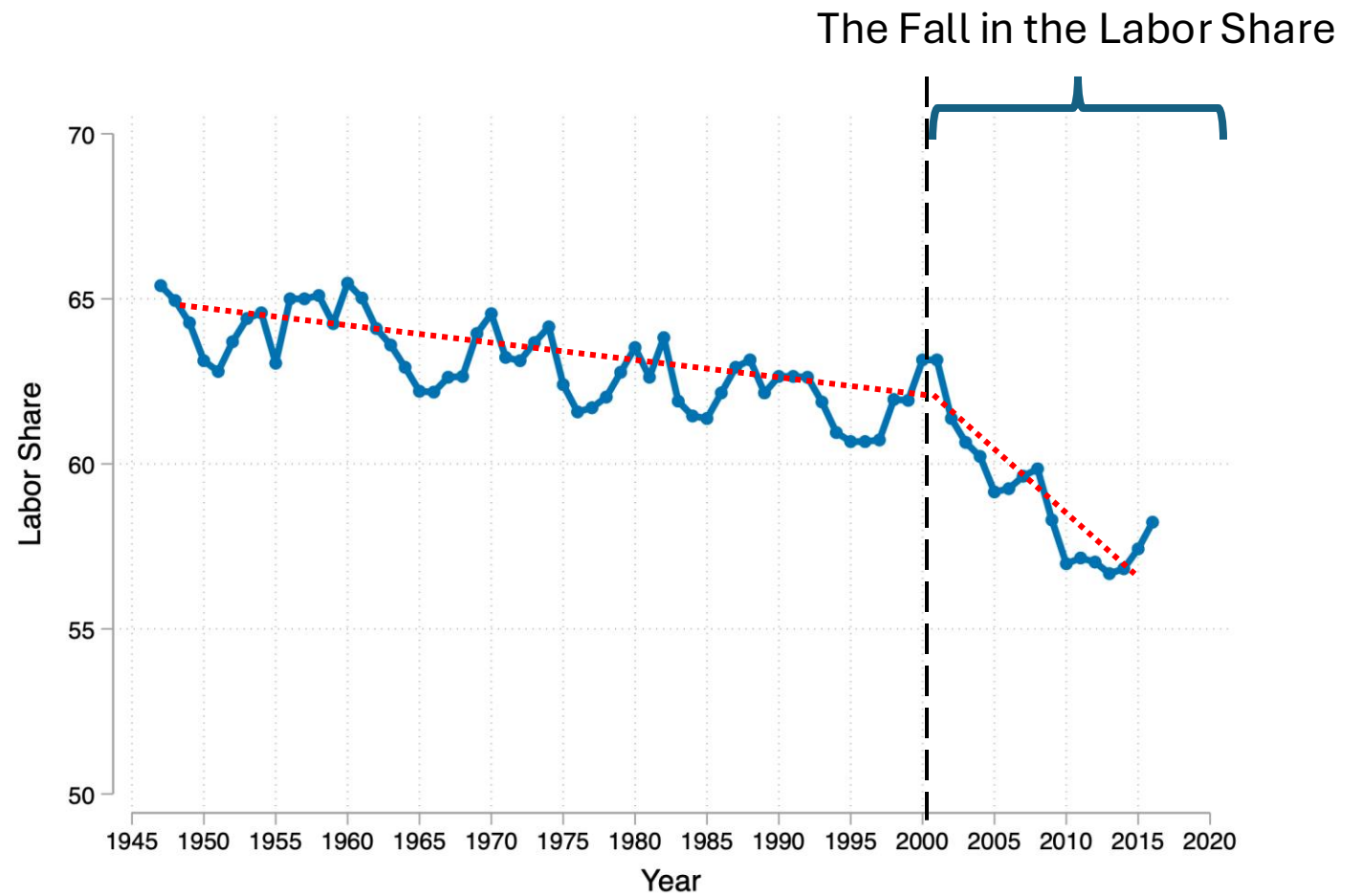
The Labor Share



The Labor Share



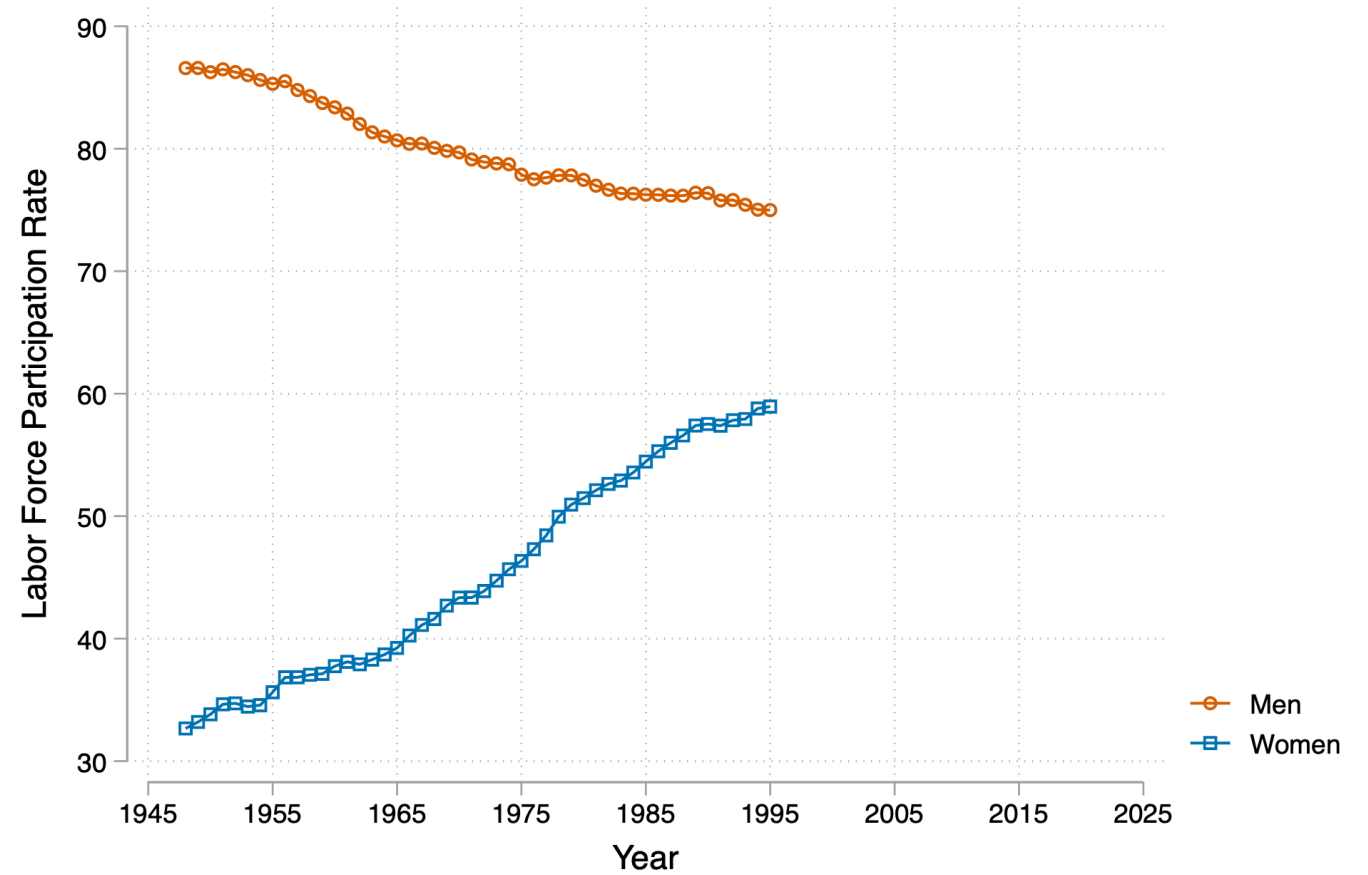
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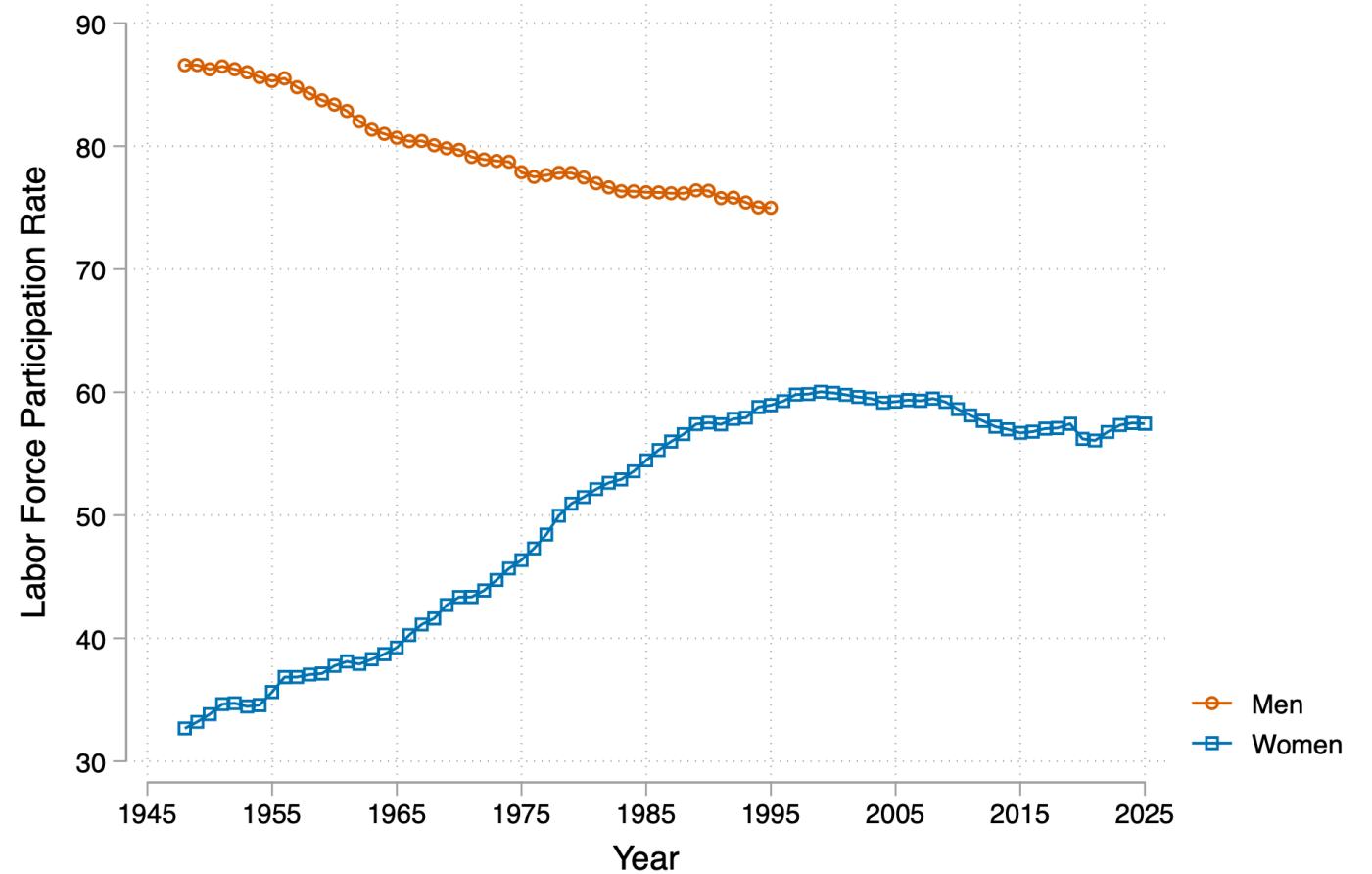
Question 3: Dramatic Shifts in Labor Force Participation

- In the news, you often see reports of fluctuating unemployment rates
 - The unemployment rate is the fraction of individuals who are actively looking for work who do not currently have a job (we will study this closely soon)
- If you don't have a job and are not actively looking for one, then you are referred to as out-of-the labor force
- What is often ignored is that there have been some pretty drastic changes in labor force participation rates over the last 75 years

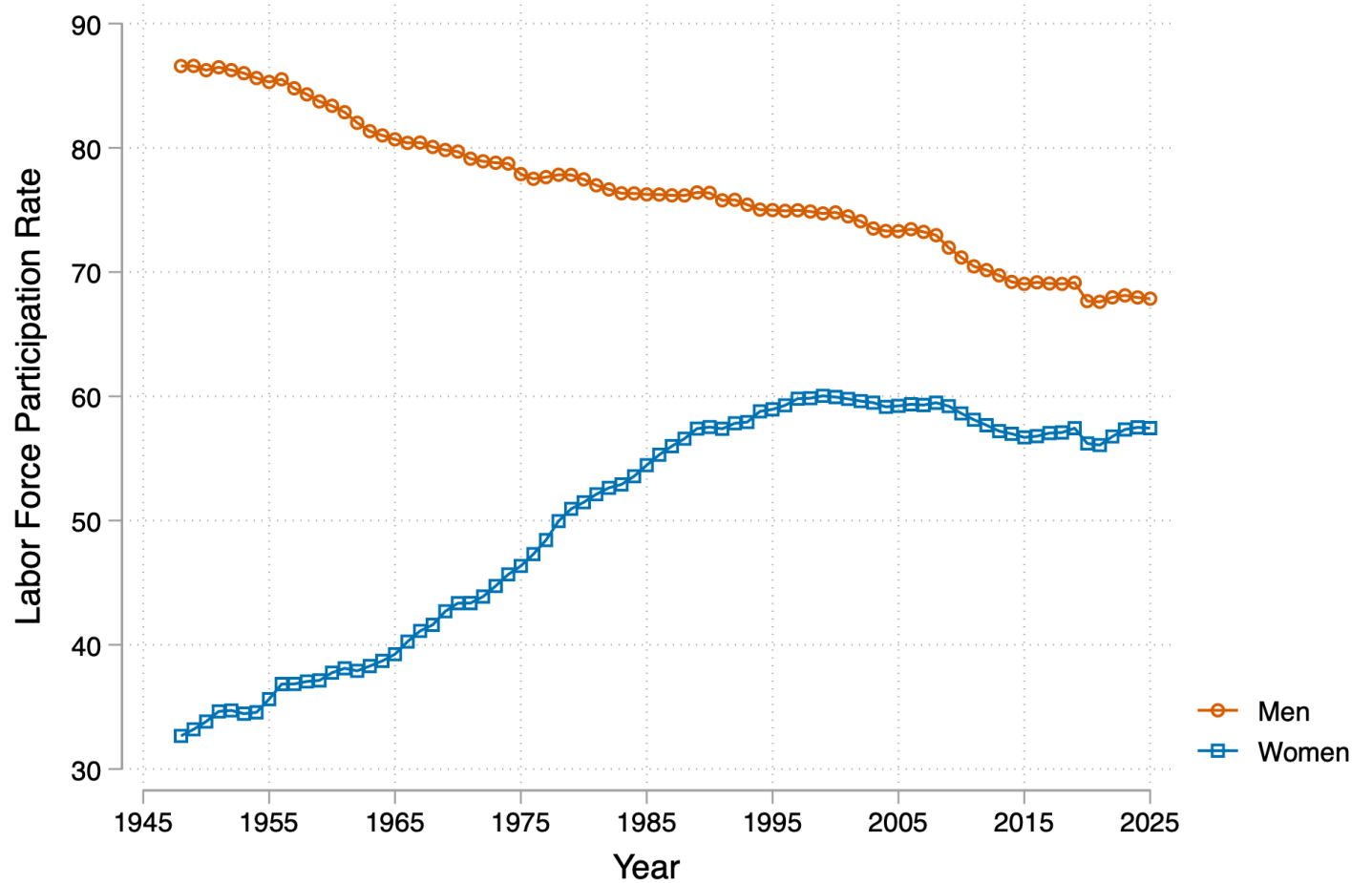
Labor Force Participation Rates



Labor Force Participation Rates



Labor Force Participation Rates



Explanations

- All of these trends boil down to what determines who works and how much they get paid?
- We will study three broad classes of explanations
 - Technology – how has the rise of computers changed jobs
 - Competition – how has rise of competition among foreign firms as well as competition for workers between domestic firms impacted the labor market
 - Institutions – how have institutions such as the minimum wage, employment laws, and unions impacted the wage structure

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Economic Models

- To guide our analysis we will consider a few different models of the labor market
- In reality, the labor market is an extremely complex system
- But models can be useful as a way to guide analysis and make sure we are internally consistent with our arguments
 - As we will see clearly with the minimum wage, political debates about the minimum wage can be reframed as debates on what is the best way to model the labor market



Are there any concepts from this set of slides that are unclear?